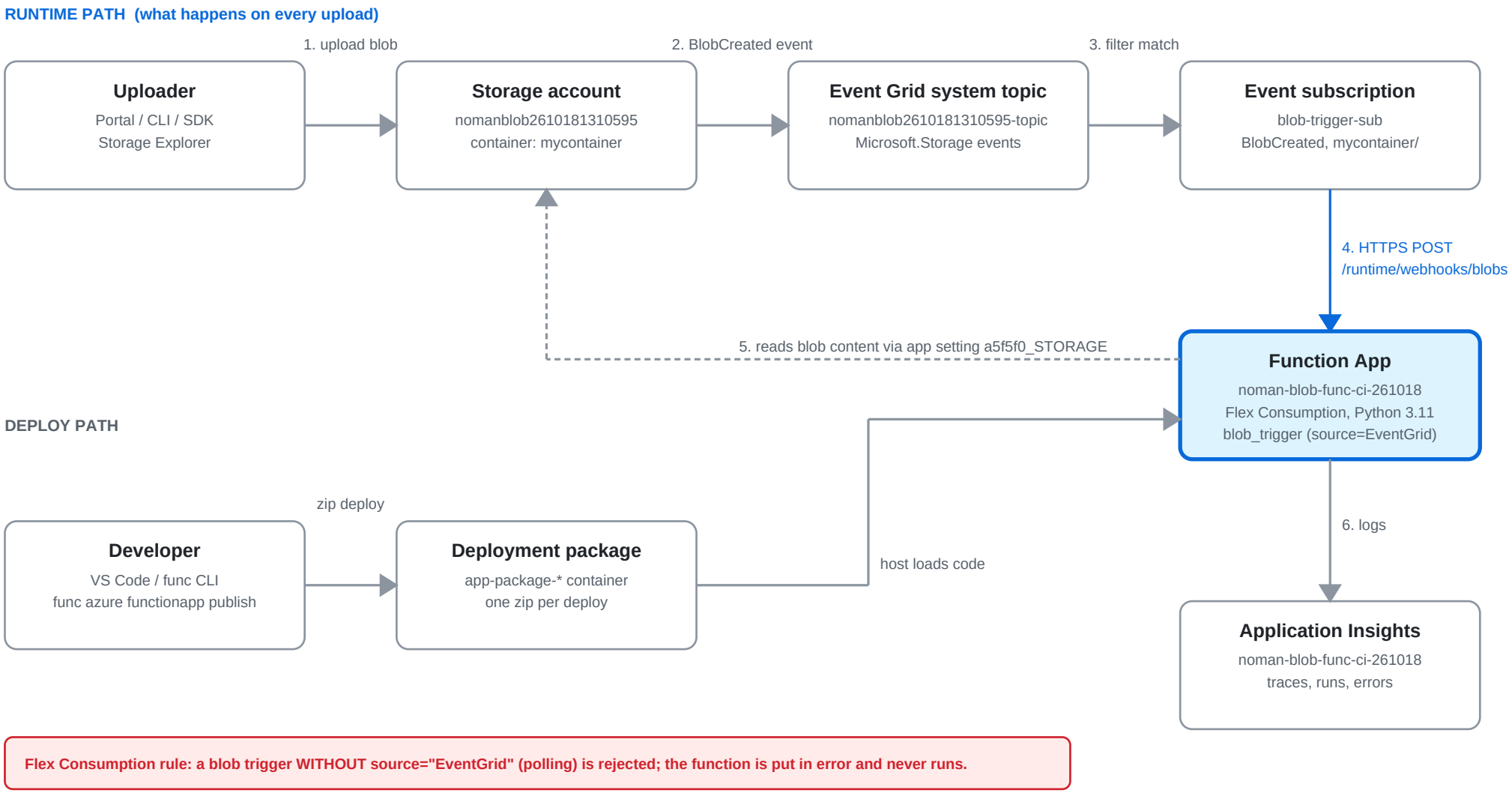
Azure Blob Trigger - Architecture

A Python Azure Function on Flex Consumption that runs on every file uploaded to **mycontainer**, driven by Event Grid. Resource group: **noman-blob-rg-2610181310595**.



Azure resources

Resource	Name	Role
Resource group	noman-blob-rg-2610181310595	Holds every resource below
Storage account	nomanblob2610181310595	Uploads (mycontainer), host state, deployment packages
Function App	noman-blob-func-ci-261018	Linux, Python 3.11, Flex Consumption plan
App Service plan	ASP-nomanblobrg2610181310595-69c0	Flex Consumption plan
Event Grid system topic	nomanblob2610181310595-topic	Publishes Microsoft.Storage events
Event Grid subscription	blob-trigger-sub	BlobCreated, subject begins with /blobServices/default/containers/mycontainer/, webhook to the function
Application Insights	noman-blob-func-ci-261018	Logs, run history, errors

Request flow

Step	What happens
1	A file is uploaded to the **mycontainer** container.
2	Storage publishes a **Microsoft.Storage.BlobCreated** event to the system topic.
3	Subscription **blob-trigger-sub** matches the event type and the mycontainer/ subject prefix.
4	Event Grid POSTs the event to **/runtime/webhooks/blobs?functionName=Host.Functions.blob_trigger**, authenticated with the blobs_extension system key. The Flex app wakes from zero if needed.
5	The host reads the blob using the **a5f5f0_STORAGE** app setting and runs **blob_trigger**.
6	The function logs the blob name and size to Application Insights (1-3 minutes ingestion delay).

Key configuration

Item	Value	Why it matters
Trigger source	source="EventGrid"	Mandatory on Flex Consumption; polling is rejected
Trigger path	mycontainer/{name}	Only blobs in mycontainer match
Connection	a5f5f0_STORAGE (app setting)	local.settings.json is not deployed; must exist in Azure
Extension bundle	[4.*, 5.0.0)	Includes the Storage Blobs extension with Event Grid support
Event coverage	Uploads after the subscription exists	Event Grid never replays older uploads
Delivery retry	30 attempts over 24 hours	Undelivered events are dropped after that